OIS-002

Canonical Model

1. Purpose

This specification defines the canonical object model required by every Organodynamics-compatible implementation.
The model describes constitutional concepts.
It SHALL NOT prescribe storage technology, programming language, network topology, database architecture, or user interface.

2. Canonical Layers

Every implementation SHALL expose the following constitutional layers.

Constitution
↓
Canon
↓
Places
↓
Experiments
↓
Contributions
↓
Evidence
↓
Understanding

These layers SHALL remain conceptually distinguishable.
An implementation MAY merge them internally.
An implementation SHALL preserve their constitutional semantics.

3. Layer Definitions

Constitution

Purpose
Defines immutable constitutional principles.
Properties
Immutable during runtime.
Globally identifiable.
Versioned.
Auditable.
Operations
Load
Verify
Upgrade (constitutional process only)
Runtime SHALL NOT modify Constitution.

Canon

Purpose
Defines vocabulary, entities, relationships and protocols.
Properties
Versioned.
Traceable.
Extensible.
Operations
Lookup
Reference
Extend
Deprecate

Place

Purpose
Persistent environment where understanding evolves.
Lifecycle
Created
Active
Dormant
Archived
Recovered
Destroyed (constitutional authorization only)

Experiment

Purpose
Structured learning process.
Lifecycle
Created
Running
Paused
Completed
Superseded
Archived

Contribution

Purpose
Intentional addition affecting understanding.
Lifecycle
Draft
Published
Referenced
Deprecated
Archived

Evidence
Purpose
Influence confidence.
Evidence SHALL NEVER directly modify constitutional truth.
Evidence MAY influence:
Defaults
Confidence
Priorities
Future experiments

4. Relationships
The following relationships are constitutional.

Place
 HAS Participants

Participant
 CREATES Contributions

Contribution
 SUPPORTS Experiment

Experiment
 PRODUCES Evidence

Evidence
 MODIFIES Confidence

Confidence
 INFLUENCES Decisions

Decisions
 NEVER modify Constitution

5. Constitutional Identity
Every canonical object SHALL possess:
Persistent Identifier
Creation Timestamp
Origin
Current Lifecycle State
Version
History
Relationships
Identity SHALL survive implementation migration.

6. Runtime Independence

The following SHALL NOT affect constitutional validity:
Programming Language
Database
Operating System
Frontend Framework
Cloud Provider
AI Provider
Deployment Model
Communication Protocol
Reference implementation MAY recommend technologies.
The specification SHALL remain technology-independent.

7. Failure Recovery

Every implementation SHALL survive:
UI replacement
Database migration
Network partition
AI replacement
Builder replacement
Repository migration
Place continuity SHALL remain reconstructable.

8. Builder Responsibility

The Builder SHALL implement behavior.
The Builder SHALL NOT reinterpret constitutional meaning.
If ambiguity is discovered:
The implementation SHALL preserve existing constitutional behavior.
The ambiguity SHALL be reported as a constitutional proposal rather than silently resolved.

9. Compliance Levels

Level 0
Canonical Reader
May read Constitution.
Cannot modify runtime.

Level 1
Builder
Can create compatible Places.

Level 2
Maintainer
Can evolve implementation.
Cannot evolve Constitution.

Level 3
Constitution Steward
May participate in constitutional evolution.
Requires constitutional process.

10. Reference Validation

A Place SHALL be considered compliant if:
All constitutional invariants hold.
All required behaviors exist.
All lifecycle rules are respected.
All conformance tests succeed.
Independent implementations exhibit equivalent constitutional behavior.